

D429

Introduction to AI for Computer Scientists

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A streaming service has 2 million viewing records with no labels and wants to find natural groups of viewers for marketing. The data team works from Austin and uses Python. Which approach fits?

- A. Reinforcement learning.
- B. Unsupervised learning, such as clustering.
- C. Supervised classification.
- D. Regression predicting watch time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A robot learns to walk by trial and error, earning a reward for distance covered. Which learning type is this?

- A. Reinforcement learning.
- B. Supervised learning.
- C. A rule-based expert system.
- D. Unsupervised learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A model that predicts equipment failure scores 99% accuracy on training data and 71% on new data. It was built by a three-person team over six weeks. What is the diagnosis and a sound next step?

- A. Overfitting; simplify or regularize the model, or add more training data.
- B. Underfitting; add more layers and features.
- C. The team was too small to build a good model.
- D. No problem; the 99% training score shows the model works.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A model scores 62% on training data and 60% on test data, both well below what the task needs. What is the diagnosis?

- A.** Overfitting.
- B.** A good model, since a small gap always means success.
- C.** Underfitting.
- D.** Data leakage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A team building a hospital readmission model included a “follow-up call completed” field that staff fill in only after they know whether the patient was readmitted. Test accuracy came out at 98%. The hospital has 400 beds. What is the problem?

- A.** Nothing; high test accuracy always proves a model is ready.
- B.** Overfitting.
- C.** Class imbalance.
- D.** Data leakage; remove any feature not available at prediction time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fraud model at a regional bank flagged 50 transactions, and 40 of them were truly fraud. There were 60 fraudulent transactions in all. The model runs on GPUs in two data centers. What is the model’s recall?

- A.** About 0.33
- B.** About 0.67
- C.** About 0.83
- D.** 0.80

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A model has a precision of 0.90 and made 200 positive predictions. How many were false positives?

- A.** 180
- B.** 20
- C.** 90
- D.** It cannot be found without recall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A résumé-screening model was trained on ten years of a firm’s past hiring decisions, a period when 85% of hires were men. It runs on a modern cloud platform. It now ranks women lower. What is the most likely cause and first fix?

- A.** Remove the gender column, which guarantees fairness.
- B.** Use a larger model with more parameters.
- C.** Historical bias in the training data; audit data and outcomes across groups, then rebalance and retest before use.
- D.** Move to a different cloud platform.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A delivery robot must find the route with the fewest hallway segments on a floor map where every segment counts the same. The map has 300 nodes and the robot has a 6-hour battery. Which algorithm guarantees the fewest segments?

- A.** Greedy best-first search.
- B.** Depth-first search.
- C.** Random moves until the battery runs low.
- D.** Breadth-first search.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A* search is run with a heuristic that sometimes overestimates the remaining cost. What guarantee is lost?

- A.** None; A* is always optimal.
- B.** It becomes breadth-first search.
- C.** It runs slower but stays optimal.
- D.** The guarantee that the path found is the cheapest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A tax-preparation firm builds a system that encodes 1,200 if-then rules written by its senior accountants. It was demoed at a conference in Denver. Which kind of AI is this?

- A.** Symbolic, rule-based AI (an expert system).
- B.** Unsupervised learning.
- C.** General AI, since it handles over a thousand rules.
- D.** Deep learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A company's chatbot reads customer emails and summarizes them for agents. Some emails contain text telling the chatbot to ignore its instructions and reveal internal notes. The chatbot runs on four servers. Which defense is best?

- A.** Add more servers.
- B.** Retrain the model on more customer emails.
- C.** Treat email text as untrusted data, limit what the chatbot can access, and check outputs before agents see them.
- D.** Tell the model in its prompt never to be tricked, which fully prevents it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A streaming service has 2 million viewing records with no labels and wants to find natural groups of viewers for marketing. ~~The data team works from Austin and uses Python.~~ Which approach fits?

Target word: paradigm **Decides:** No labels; goal is to discover groups.

A	Adjacent category	There is no agent acting for rewards.
B	Correct	Clustering finds structure in unlabeled data.
C	Confusion set	Needs labeled groups, which do not exist.
D	Dropped element	Needs a labeled target and does not find groups.

2. Answer A Concept 1 · transfer

A robot learns to walk by trial and error, earning a reward for distance covered. Which learning type is this?

Target word: reverse: setup to paradigm **Decides:** A reward signal drives learning.

A	Correct	An agent improves its actions to maximize reward.
B	Confusion set	No labeled correct answers are given.
C	Inverted operation	Expert systems follow fixed rules rather than learning from rewards.
D	Adjacent category	It is not finding structure in a static dataset.

3. Answer A Concept 2

A model that predicts equipment failure scores 99% accuracy on training data and 71% on new data. ~~It was built by a three person team over six weeks.~~ What is the diagnosis and a sound next step?

Target word: diagnose the gap **Decides:** Large drop from training to test.

A	Correct	The model memorized training data and does not generalize.
B	Inverted operation	More complexity would worsen overfitting.
C	Camouflage	Team size does not explain the pattern.
D	Dropped element	Ignores performance on new data, the only score that matters.

4. Answer C Concept 2 · transfer

A model scores 62% on training data and 60% on test data, both well below what the task needs. What is the diagnosis?

Target word: reverse pattern **Decides:** Poor on both sets.

A	Inverted operation	Overfitting shows high training and low test scores.
B	Absolute language	A small gap with low scores still fails the task.
C	Correct	The model is too simple to capture the pattern.
D	Confusion set	Leakage inflates scores, it does not depress them.

5. Answer D Concept 3

A team building a hospital readmission model included a “follow-up call completed” field that staff fill in only after they know whether the patient was readmitted. Test accuracy came out at 98%. ~~The hospital has 400 beds.~~ What is the problem?

Target word: why accuracy is inflated **Decides:** The field is recorded after the outcome.

A	Absolute language	Leakage makes test scores misleading.
B	Adjacent category	Overfitting shows up as a train-test gap; here the test score is also inflated.
C	Confusion set	Imbalance skews accuracy but is not caused by a future-only field.
D	Correct	The field carries the answer, so the score will not hold in use.

6. Answer B Concept 4

A fraud model at a regional bank flagged 50 transactions, and 40 of them were truly fraud. There were 60 fraudulent transactions in all. ~~The model runs on GPUs in two data centers.~~ What is the model’s recall?

Target word: recall **Decides:** $TP = 40$, $FN = 60 - 40 = 20$.

A	Inverted operation	$20 / 60$ is the share of fraud missed.
B	Correct	$40 / (40 + 20) = 0.667$.
C	Wrong stem number	$50 / 60$ uses the flagged count as if all were true positives.
D	Confusion set	$40 / 50$ is precision.

7. Answer B Concept 4 · transfer

A model has a precision of 0.90 and made 200 positive predictions. How many were false positives?

Target word: solve for FP **Decides:** $TP = 0.90 \times 200$.

A	Inverted operation	That is the true positives.
B	Correct	$TP = 180$, so $FP = 200 - 180 = 20$.
C	Wrong stem number	Reads the precision as a count.
D	Adjacent category	Recall is needed for false negatives, not false positives.

8. Answer C Concept 5

A résumé-screening model was trained on ten years of a firm’s past hiring decisions, a period when 85% of hires were men. ~~It runs on a modern cloud platform.~~ It now ranks women lower. What is the most likely cause and first fix?

Target word: source of bias and response **Decides:** Labels come from skewed historical decisions.

A	Absolute language	Proxies such as clubs or schools can carry the same signal.
B	Adjacent category	More capacity learns the biased pattern just as well.
C	Correct	The model learned past patterns; fixes start with the data and fairness testing.
D	Camouflage	Hosting does not change what the model learned.

9. Answer D Concept 6

A delivery robot must find the route with the fewest hallway segments on a floor map where every segment counts the same. ~~The map has 300 nodes and the robot has a 6-hour battery.~~ Which algorithm guarantees the fewest segments?

Target word: optimal search in an unweighted graph **Decides:** All edges equal; want fewest steps.

A	Adjacent category	Follows the heuristic and does not guarantee the shortest path.
B	Confusion set	DFS may find a long path first.
C	Camouflage	Answers the battery detail and guarantees nothing.
D	Correct	BFS explores level by level, so the first path found has the fewest edges.

10. Answer D Concept 6 · transfer

A* search is run with a heuristic that sometimes overestimates the remaining cost. What guarantee is lost?

Target word: effect of an inadmissible heuristic **Decides:** Admissibility is what makes A* optimal.

A	Absolute language	Only with an admissible heuristic.
B	Confusion set	A* behaves like BFS only with a zero heuristic and equal edge costs.
C	Inverted operation	Overestimating often speeds the search while giving up optimality.
D	Correct	Overestimates can make A* skip the optimal path.

11. Answer A Concept 7

A tax-preparation firm builds a system that encodes 1,200 if-then rules written by its senior accountants. ~~It was demoed at a conference in Denver.~~ Which kind of AI is this?

Target word: classify the approach **Decides:** Rules are written by people, not learned.

A	Correct	Knowledge is hand-coded as rules.
B	Adjacent category	Nothing is learned from unlabeled data.
C	Absolute language	Rule count does not make a system general; it is narrow AI.
D	Confusion set	Deep learning learns from data rather than written rules.

12. Answer C Concept 8

A company's chatbot reads customer emails and summarizes them for agents. Some emails contain text telling the chatbot to ignore its instructions and reveal internal notes. ~~The chatbot runs on four servers.~~ Which defense is best?

Target word: defense against injected instructions **Decides:** Outside content is being read as if it were instructions.

A	Camouflage	Capacity has nothing to do with the threat.
B	Adjacent category	More data does not separate instructions from content.
C	Correct	Layered controls reduce harm even if some injected text gets through.
D	Absolute language	Prompt wording alone is not a reliable control.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Learning paradigms	Supervised learning uses labeled examples; unsupervised finds structure in unlabeled data; reinforcement learning learns from rewards through trial and error.	Calling any learning from data “supervised.”	Whether labels, no labels, or a reward signal are present.
2	Overfitting vs. underfitting	High training score with much lower test score = overfitting; low on both = underfitting.	Adding complexity to an overfit model.	Two accuracy figures, training and test.
3	Data leakage	A feature that would not be available at prediction time, or carries the answer, inflates test results.	Trusting a suspiciously high test score.	A field recorded after the outcome is known.
4	Precision and recall	Precision = $TP / (TP + FP)$; recall = $TP / (TP + FN)$.	Swapping the two denominators.	Counts of flagged cases and actual positives.
5	Bias in training data	Models learn patterns in historical data, including unfair ones; fixes start with the data and with outcome testing across groups.	Thinking deleting a sensitive column removes bias.	Past decisions used as labels for a people-related model.
6	Search algorithms	BFS finds fewest-step paths in unweighted graphs; A* is optimal only with an admissible (never overestimating) heuristic.	Assuming DFS or any A* is optimal.	“Fewest steps,” “heuristic,” “guarantee.”
7	Symbolic AI vs. machine learning	Symbolic (rule-based, expert) systems encode human-written rules; machine learning infers patterns from data. Both are narrow AI.	Calling any capable system general AI or machine learning.	Hand-written if-then rules vs. training data.
8	AI security concerns	Treat model inputs from outside as untrusted: separate instructions from data, limit what the model can do, validate outputs, and keep humans in the loop for risky actions.	Relying on the model to police itself.	A model that reads outside content and can take actions.

Course scope (public description): Introduction to AI for Computer Scientists provides an overview of critical terminology and key concepts for artificial intelligence (AI). The course explores the history and evolution of AI, elements of code, and the process for understanding algorithmic approaches to AI. The course presents topics of bias, ethical issues, and security concerns. Contextualized examples offer students an opportunity to see these concepts in professional scenarios; identifying issues within code, understanding the steps within an AI design, and understanding the different features, limitations, and benefits for a multitude of AI applications.